Parasitology

Dr Ayman Ibrahim

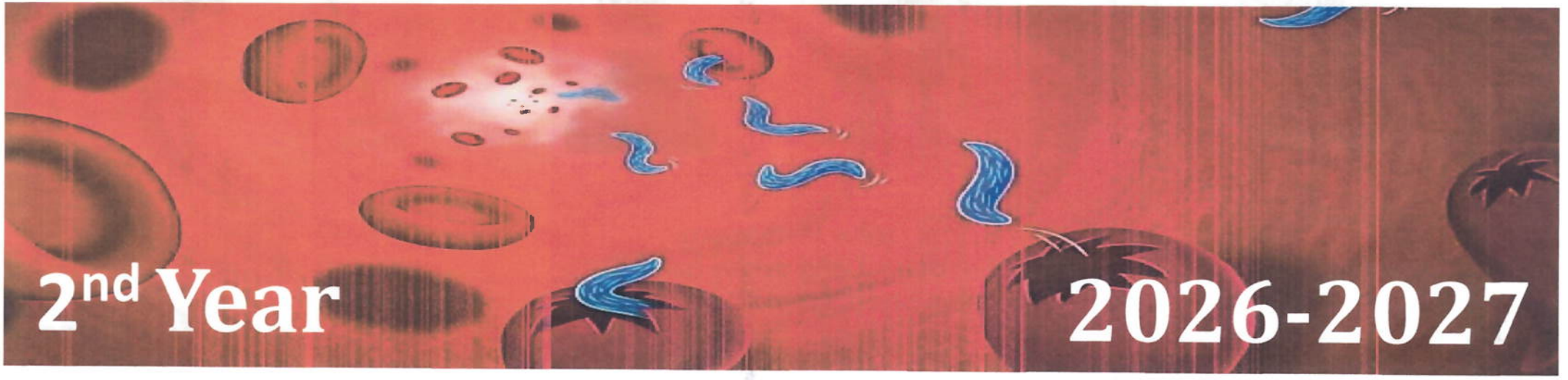




Blood

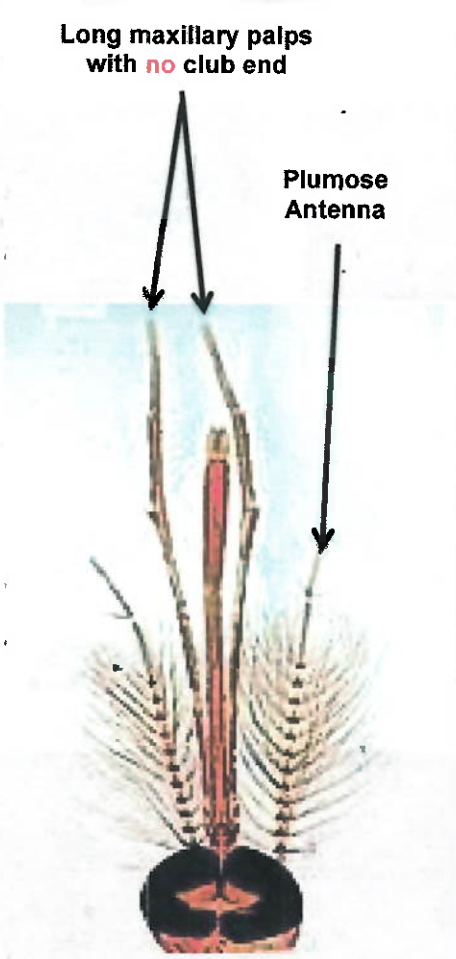
Practical

Pulse.

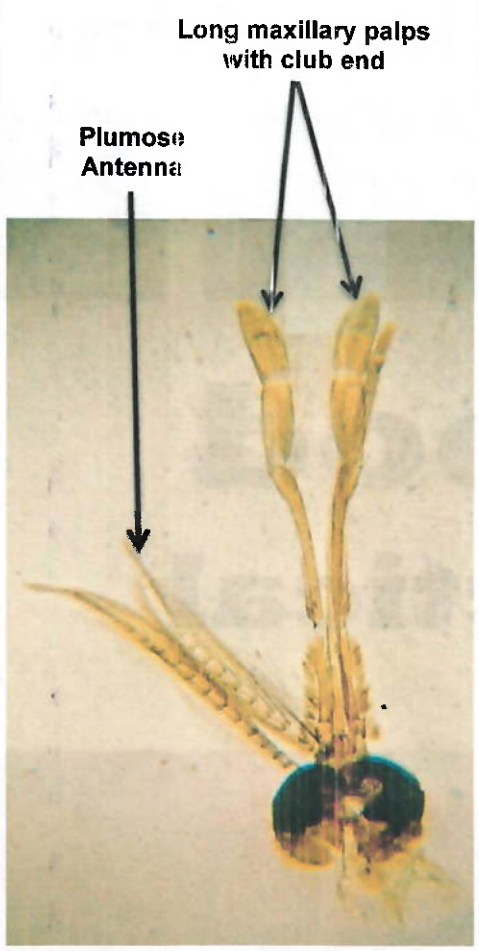


Arthropods

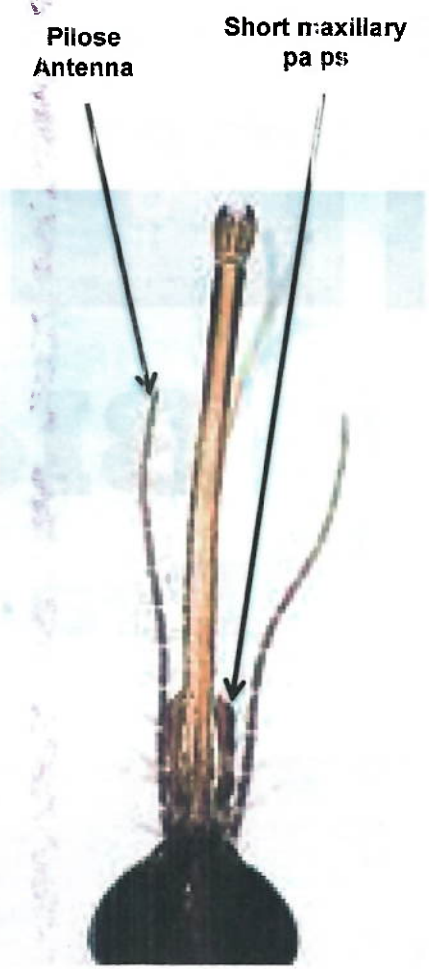
Culex Male MP



Anopheles Male MP



Culex Female MP



Anopheles Female MP



MP = Mouthparts
Plumose = Long dense hair = Male
Pilose = Short sporadic hair = Female

Pulse.

Culex Larva



Aedes Larva

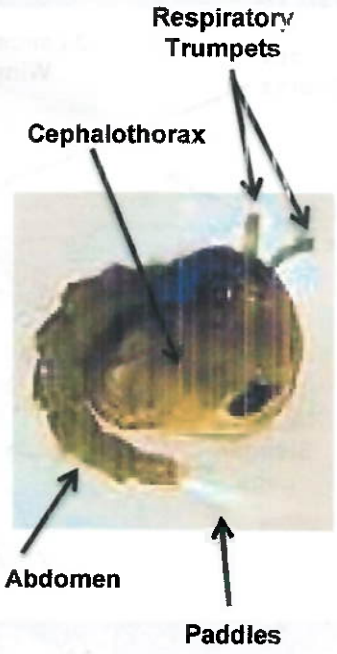


Anopheles Larva



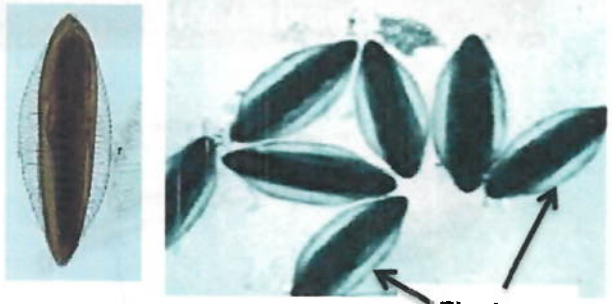
Pupa of Mosquitoes

Comma Shape



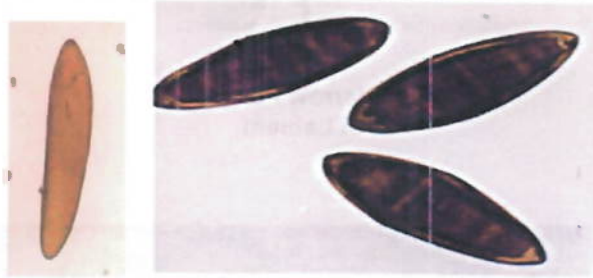


Anopheles Egg



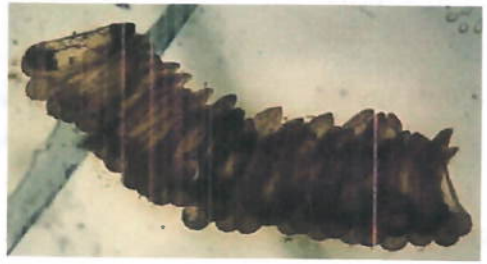
Single – Brown

Aedes Egg



Single – Brown – No Floaters

Culex Egg Raft

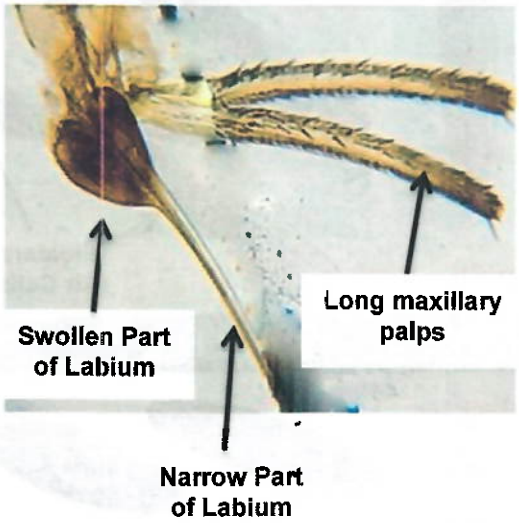


Group of eggs – Brown

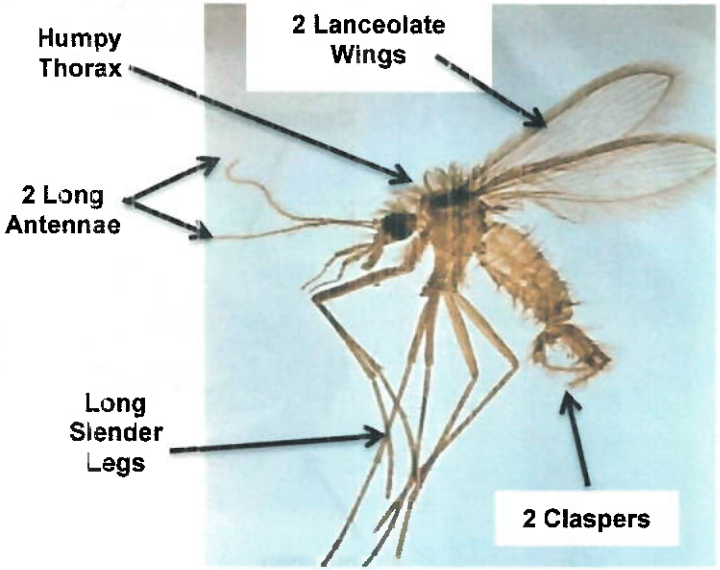


Pulse.

Glossina MP



Sand Fly Male



Sand Fly Female



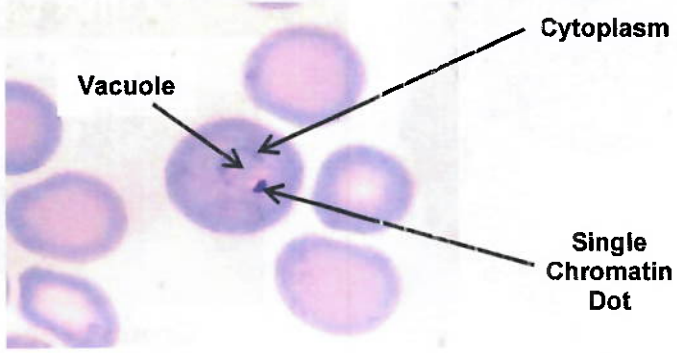
Protozoa

Plasmodium vivax Ring

1 or 2 large rings in field

Single inside RBC

With single chromatin dot

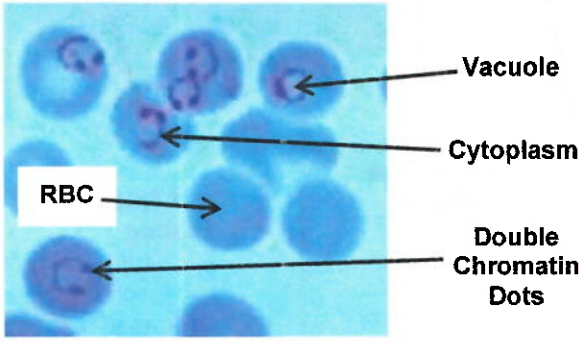


Plasmodium falciparum Ring

Large number of small rings

May be 2 rings inside one RBC

One ring may have 2 chromatin dots

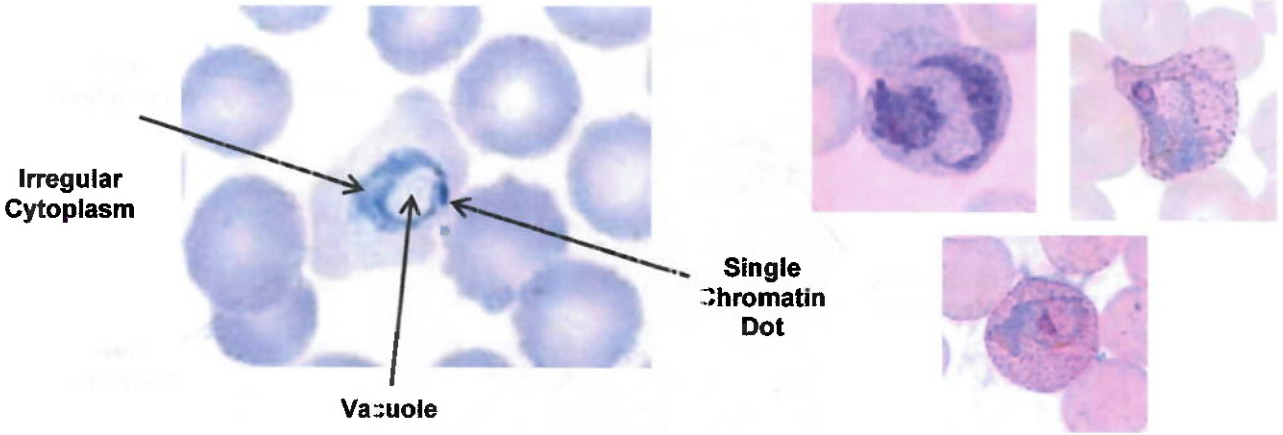




Pulse.

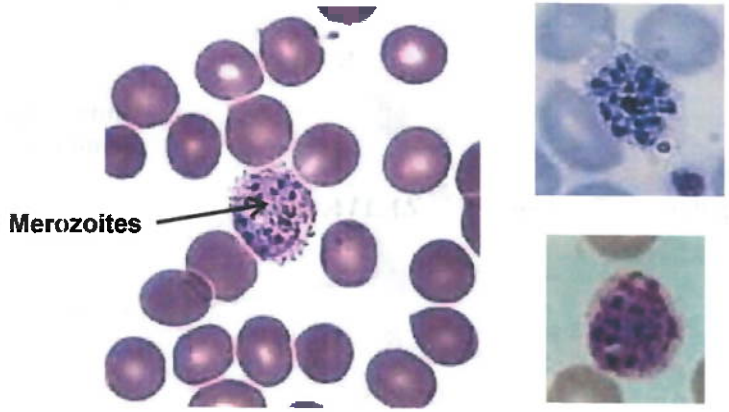
Plasmodium vivax Trophozoite

Irregular – Vacuolated – Chromatin dot



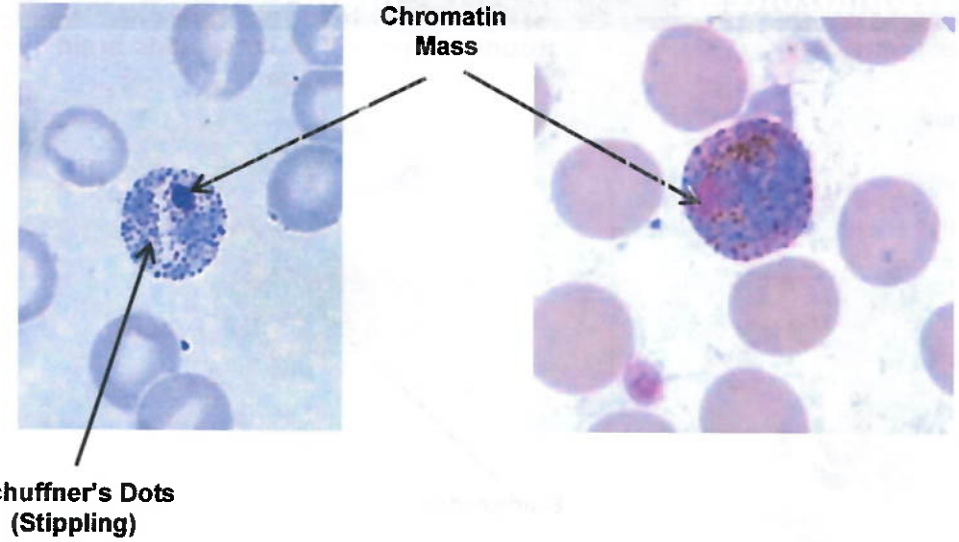
Plasmodium vivax Schizont

12-24 large dots inside RBC



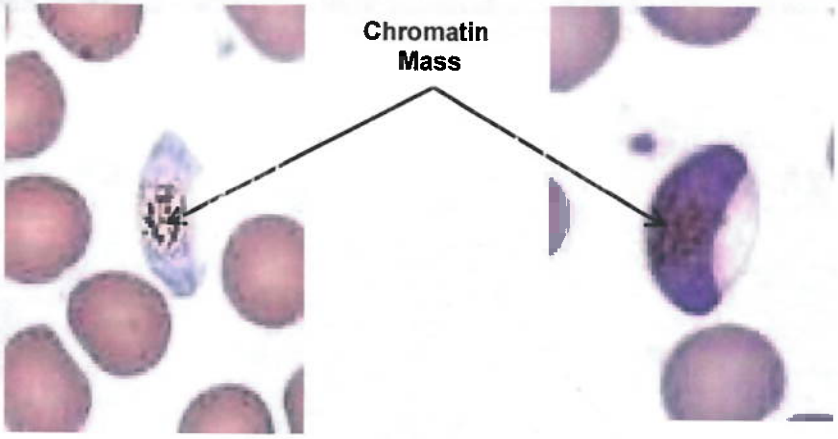
Plasmodium vivax Gametocyte

Thousands of small granules inside RBC + 1 big mass



Plasmodium falciparum Gametocyte

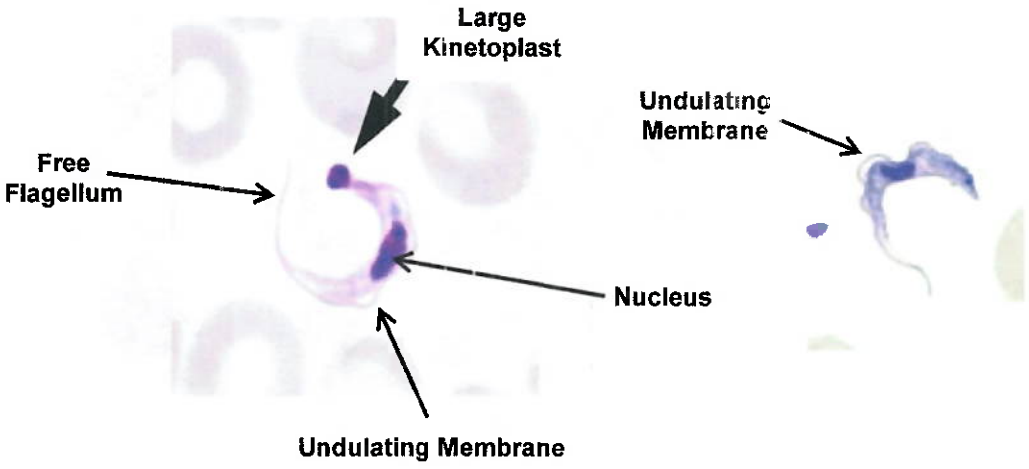
Banana-shape in blood film



Pulse.

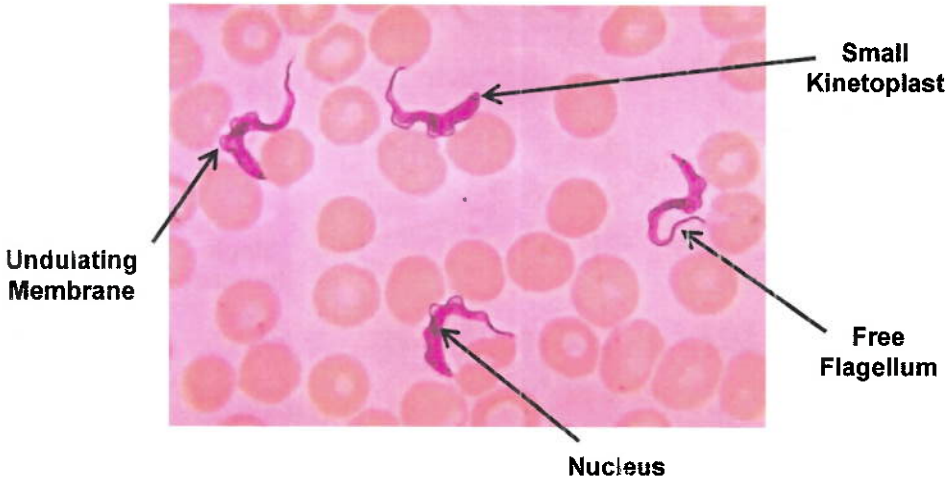
Monomorphic Trypanosoma (Trypanosoma cruzi)

C-shape – Between RBCs – One size – Large kinetoplast



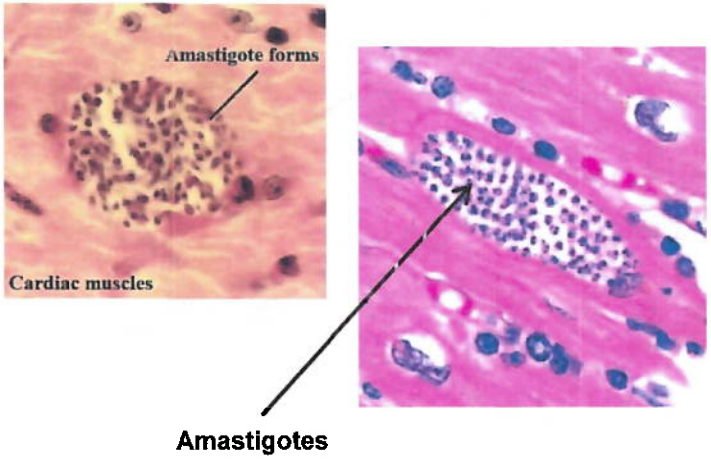
Polymorphic Trypanosoma (Trypanosoma brucei)

S or W-shape – Between RBCs – Different sizes – Small kinetoplast



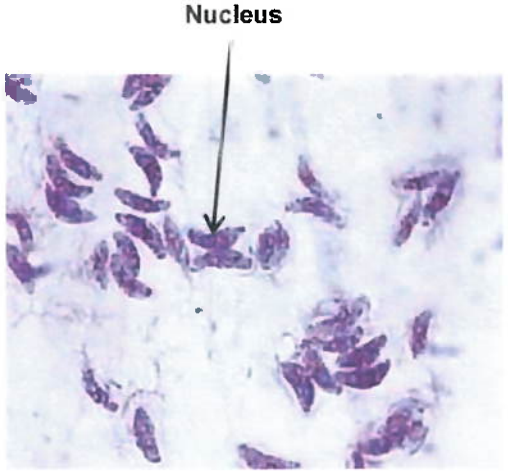
Trypanosoma cruzi Amastigote In Muscles

Collections of dots between muscles



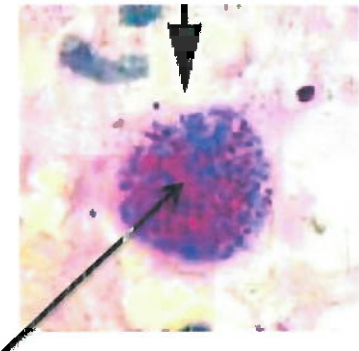
Toxoplasma gondii Tachyzoites (Trophozoites)

Small crescent parasites



Toxoplasma gondii Brain Cyst

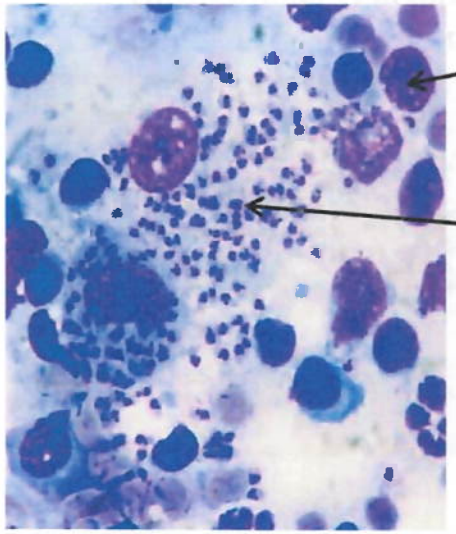
Rounded cyst full of dots inside brain tissue

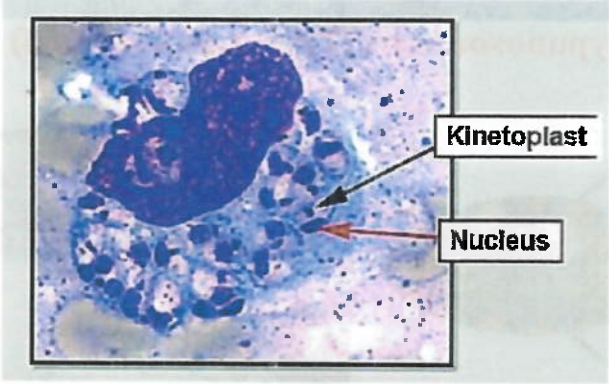


Pulse.

Amastigote of Leishmania

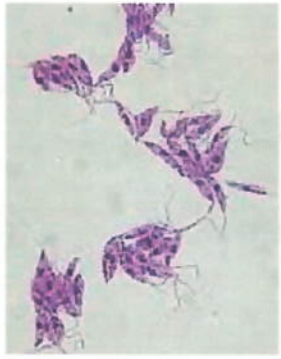
Small dots between large macrophages بنفسجي

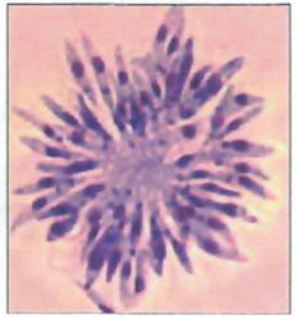


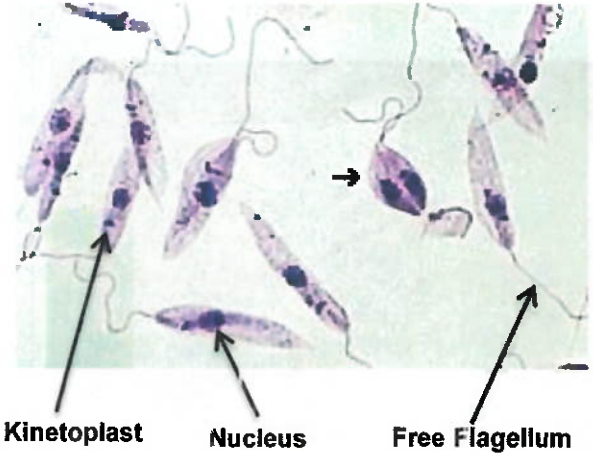


Promastigote of Leishmania

Fusiform – With free flagellum – Nucleus in the middle – Small kinetoplast at one pole

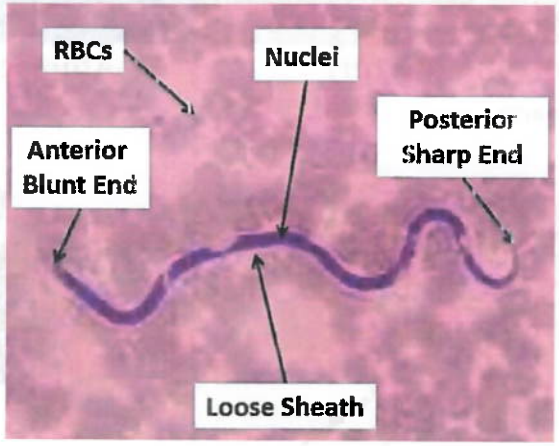


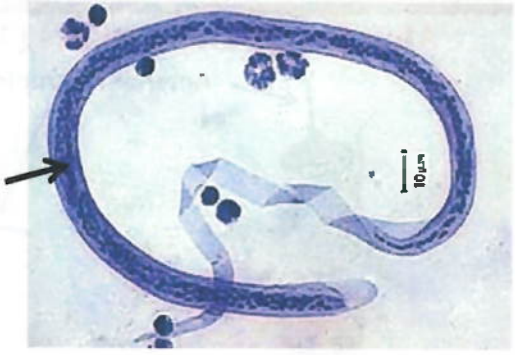




Microfilaria of Wuchereria bancrofti (Helminths)

Worm-like inside blood film





Pulse.

Adult Vectors

Sand Fly Female

Leishmania



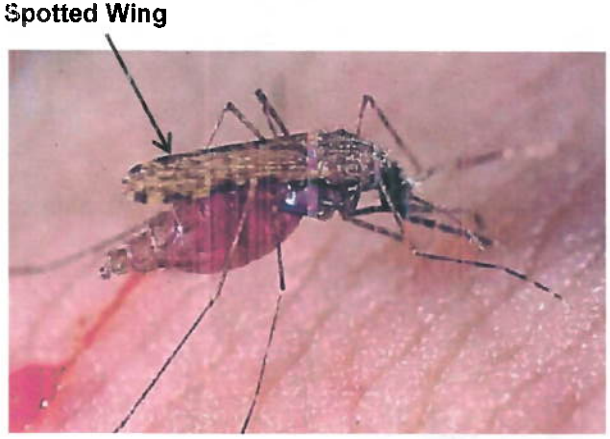
Culex Female

Wuchereria bancrofti



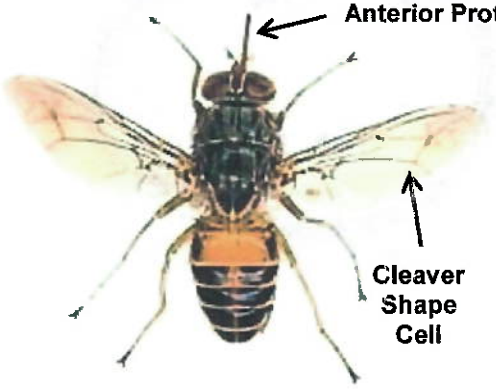
Anopheles Female

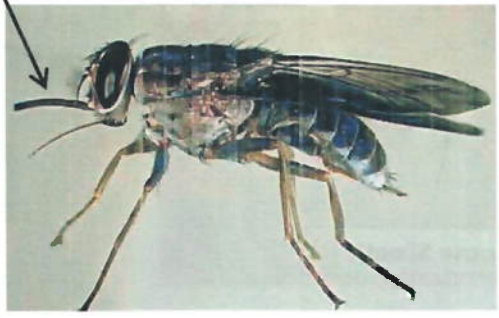
Malaria (Plasmodium)



Glossina (Tsetse Fly)

African trypanosomiasis (Trypanosoma brucei)





Triatoma megista

American trypanosomiasis (Trypanosoma cruzi)





Pulse.

Cases

Lymphatic (Bancroftian) Filariasis

Wuchereria bancrofti

Red hot tender streaks or elephantiasis in lower limbs + Passage of lymph with urine (Chyluria)

Infective stage: 3rd stage filariform larva

Diagnostic stage: Microfilaria in blood (10pm-2am)

Mode of infection: Active penetration of the skin by 3rd stage filariform larva during biting of vector

Vector: Culex mosquito female

Type of transmission: Cyclodevelopmental





Visceral Leishmaniasis (Kala azar)

Leishmania donovani

Fever + Hepatosplenomegaly + Lymphadenopathy + Pancytopenia

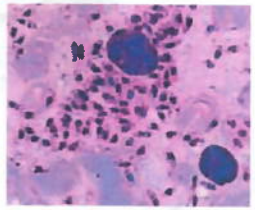
Infective stage: Promastigote

Diagnostic stages: Amastigote in tissue
Promastigote in culture

Mode of infection: Bite of sand fly female

Vector: Sand fly female (Phlebotomus – Lutzomyia)

Type of transmission: Cyclopropagative



American Trypanosomiasis (Chagas Disease)

Trypanosoma cruzi

Acute stage: Romana Sign

Chronic stage: Cardiomegaly – Megacolon
Megaoesophagus

Infective stage: Metacyclic trypomastigote

Diagnostic stage:

Acute infection: Trypomastigote in blood

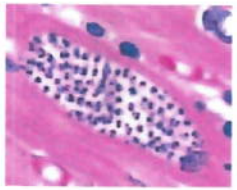
Chronic infection: Amastigote in muscles

Mode of infection: Contamination of bite wound by stool of the vector containing metacyclic trypomastigote

Type of transmission: Cyclopropagative





Pulse.

African Trypanosomiasis (Sleeping Sickness)

Trypanosoma brucei

Trypanosoma Chancre
Winterbottom's sign – Sleep disturbance

Infective stage: Metacyclic trypomastigote

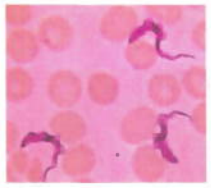
Diagnostic stage: Trypomastigote in blood & CSF

Mode of infection: Bite of Glossina

Type of transmission: Cyclopropagative







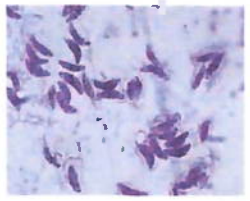
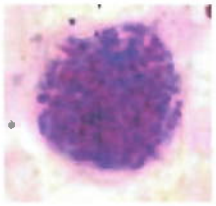
Toxoplasmosis

Toxoplasma gondii

Still birth + Hydrocephalus
(Congenital Toxoplasmosis)

Infective stages: Oocyst & Pseudocyst & True cyst

Diagnostic stages:
Tachyzoites (In acute or early infection)
Brain cyst (In chronic or latent infection)

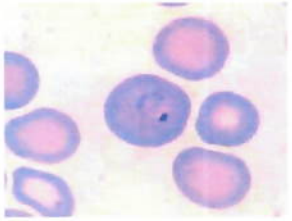
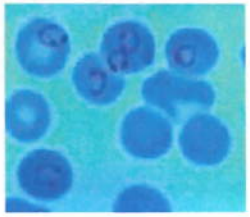




Malaria

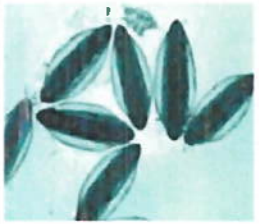
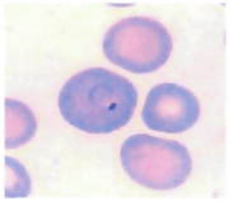
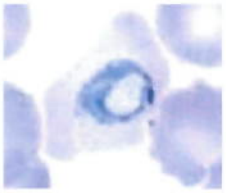
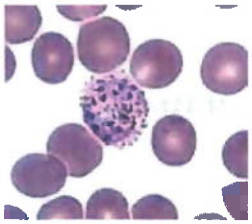
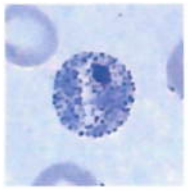
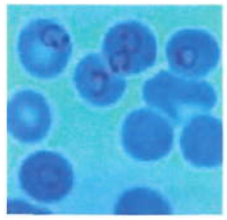
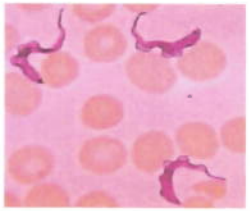
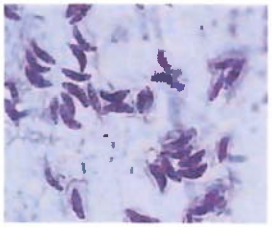
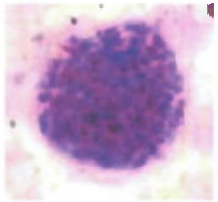
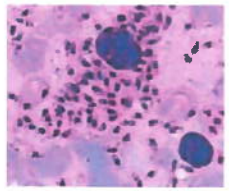
Plasmodium

Fever every 48 or 72 hours + chills + anemia + jaundice + hepatosplenomegaly

Infective stage: Sporozoite
Diagnostic stages (P. vivax): Ring - Trophozoite Schizont - Gametocyte
Diagnostic stages (P. falciparum): Ring & Gametocyte
Mode of infection: Bite of female Anopheles
Vector (DH): Female Anopheles mosquito
Type of transmission: Cyclopropagative

Pulse.

Male Anopheles MP	Male Culex MP	Female Anopheles MP	Female Culex MP	Culex Larva	Aedes Larva	Anopheles Larva
						
Pupa of Mosquito	**Anopheles Egg**	**Aedes Egg**	**Culex Egg Raft**	**Glossina MP**	**Sand Fly Male**	**Sand Fly Female**
						
W. bancrofti Microfilaria	**P. vivax Ring**	**P. vivax Trophozoite**	**P. vivax Schizont**	**P. vivax Gametocyte**	**P. falciparum Ring**	**P. falciparum Gametocyte**
						
Polymorphic Trypanosoma	**Monomorphic Trypanosoma**	**T. cruzi Amastigote**	**Toxoplasma Trophozoites**	**Toxoplasma Brain Cyst**	**Leishmania Amastigote**	**Leishmania Promastigote**
						

Pulse.